Multi-Cavity Auto Molding Encapsulation Press

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-MP-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	Multi-Cavity Auto Molding Encapsulation Press	**Document ID:**	VAI-MAN-MP-001
Machine Type Key:	molding-press	**Cleanroom Bay / Stage:**	Bay 4: Encapsulation & Mold Chase
Document Version:	1.0	**Release Date:**	2026-08-27

Automated multi-plunger encapsulation press utilizing Epoxy Molding Compound (EMC) pellets to hermetically seal wire-bonded semiconductor assemblies at 175°C under high hydraulic transfer pressure.

Process Flow: Wire-bonded leadframes are loaded into precision-machined upper and lower mold chases. Pre-heated solid EMC pellets are injected by multi-plunger hydraulic rams into runner channels, encapsulating chips without wire sweep. Degassing vacuum prevents internal micro-voiding.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Hydraulic Transfer Plunger	Compresses melted EMC resin pellets into mold cavities under 120-180 bar.	Ram pressure (bar), transfer velocity (mm/s), plunger load (kN).	Hydraulic pressure surge (>165 bar), piston seal leakage, plunger friction.
Mold Chase Heater Platens	Maintains upper and lower mold cavities at 170-180°C curing temp.	Platen temperature (°C), temperature uniformity across chase (±2°C).	Thermal gradient (>5°C), heater cartridge resistance drift.
Mold Cavity Air Vents	Allows trapped air to escape during resin transfer to prevent voids.	Vent depth (µm), vent vacuum level (kPa).	Resin flash accumulation in vents, mold void defects.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Hydraulic Pressure pressure_hydraulic	bar	120 – 150	**150 – 175**	**175 – 250**	Higher is Worse
Mold Chase Temp temperature_mold	°C	170 – 180	**180 – 192**	**192 – 250**	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 1,980 operating hours. **Model:** RUL = BaseLife * (1 - (0.50*wear_hyd_press + 0.30*wear_temp + 0.20*wear_plunger))

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Hydraulic Pressure	pressure_hydraulic	bar	**0.50**	135	185	HIGHER IS WORSE
Mold Temp	temperature_mold	°C	**0.30**	175	195	HIGHER IS WORSE
Plunger Load	load_plunger	kN	**0.20**	25	50	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	**1.00 (100%)**	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Resin Flash Bleed & Incomplete Cavity Fill SYM-MP-001	**HIGH**	pressure_hydraulic, temperature_mold	• Mold air vent blockage • Hydraulic pressure surge • Heater temperature non-uniformity	Clean mold chase vents and inspect hydraulic plunger seal rings.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: Hydraulic Transfer Ram Pressure Spike & Mold Vent Flash (SCEN-MP-001)	Severity: WARNING
Telemetry Sensor Pattern:	pressure_hydraulic > 170 bar, load_plunger > 40 kN
Possible Root Causes:	Cylinder piston ring wear | Resin flash accumulation in vents
Recommended Corrective Action:	Ultrasonic clean mold chase vents and replace hydraulic piston seal rings.
Verification & Recovery Steps:	Verify hydraulic pressure profile curve during transfer > Inspect encapsulated package cross-section for wire sweep (< 3%)